

Home Energy Dispatch and Cost Optimization

By SolarSparks

Varun E BT2024220
K. Sai Tushaar BT2024022
Tejas Kollipara BT2024147

Problem Statement

Households equipped with rooftop solar panels and battery storage still face high electricity bills due to inefficient use of available energy resources. Solar generation is intermittent, battery capacity is limited, and grid electricity prices vary throughout the day. Without intelligent control, households may charge or discharge the battery at suboptimal times, export energy when prices are low, or draw power from the grid when costs are high.

This project aims to develop an optimization-based dispatch strategy that determines the cost-minimizing hourly schedule for grid import/export, battery charging/discharging, and solar utilization over a two-month period. The model is formulated as a Linear Programming (LP) problem using real load, PV generation, and electricity price data.

The key objectives are:

- Determine hourly optimal energy flows among solar, battery, load, and the grid,
- Compute the optimal battery State-of-Charge trajectory,
- Identify the best hours to buy from or sell to the grid,
- Minimize total electricity cost over the considered period.

Dataset Overview

We used an open dataset from Kaggle and performed necessary cleaning to retain only the relevant columns:

- Timestamp,
- Demand (Load),
- Solar Generation,
- Grid Buy Price,
- Grid Sell Price.

Model Assumptions

Optimization Model Assumptions

- Pure linear program (no binary variables),
- Convergence graphs are simulated since the Pulp CBC solver does not expose iteration statistics.

Battery-Related Assumptions

Battery Capacity and Power Limits

- Total usable energy $E_{\max} = 10$ kWh,
- Max charge power $P_{\max}^{ch} = 3.5$ kW,
- Max discharge power $P_{\max}^{dis} = 3.5$ kW.

Efficiencies

- Charging efficiency $\eta_{ch} = 0.95$,
- Discharging efficiency $\eta_{dis} = 0.95$.

SOC Limits

- $SOC_{\min} = 10\% = 1.0$ kWh,
- $SOC_{\max} = 95\% = 9.5$ kWh,
- Initial SOC: $SOC_{init} = SOC_{\min}$.

Baseline and Heuristic Approaches

1. Passive Solar Model (No Battery)

PV directly serves the load; excess solar is exported to the grid. This resulted in a high total cost of ≈ 8055 rupees.

2. Smart Heuristic Strategy

1. Serve load from PV first,
2. Charge the battery with remaining PV,
3. Export leftover PV to the grid,
4. Discharge the battery during peak-price hours,
5. Import from the grid if required,
6. Maintain SOC within limits.

This reduced the cost to ≈ 5255.59 rupees. However, the heuristic does not exploit grid price variations optimally.

Linear Programming Formulation

Decision Variables

$$\begin{aligned} P_t^{buy} &\geq 0, & P_t^{sell} &\geq 0, \\ P_t^{ch} &\in [0, P_{\max}^{ch}], & P_t^{dis} &\in [0, P_{\max}^{dis}], \\ E_t &\in [SOC_{\min}, SOC_{\max}]. \end{aligned}$$

Objective Function

$$\min J = \sum_{t=1}^T \left(p_t^{buy} P_t^{buy} - p_t^{sell} P_t^{sell} + k(P_t^{ch} + P_t^{dis}) \right),$$

where

$$k = \text{DEGRAD_COST} + \text{CH_DIS_PENALTY}.$$

Constraints

$$PV_t + P_t^{buy} + P_t^{dis} = P_t^{load} + P_t^{ch} + P_t^{sell}$$

$$E_{t+1} = E_t + \eta_{ch} P_t^{ch} - \frac{1}{\eta_{dis}} P_t^{dis}$$

$$SOC_{min} \leq E_t \leq SOC_{max}$$

$$0 \leq P_t^{ch} \leq P_{max}^{ch}, \quad 0 \leq P_t^{dis} \leq P_{max}^{dis}$$

$$E_0 = SOC_{init}, \quad E_T = SOC_{init}$$

LP Model Results Overview

```
=== Optimal hourly power flow (first 12 rows) ===
```

	Timestamp	Load_kw	Solar_kw	Solar_utilized_kw	Grid_Import_kw	Grid_Export_kw	Battery_Charge_kw	Battery_Discharge_kw	Battery_Net_kw	SOC_kWh
0	2025-01-01 00:00:00	1.2322	0.0000	0.0000	1.2322	0.0	0.0000	0.0	0.0000	1.0000
1	2025-01-01 01:00:00	1.1317	0.0000	0.0000	1.1317	0.0	0.0000	0.0	0.0000	1.0000
2	2025-01-01 02:00:00	1.2576	0.0000	0.0000	1.2576	0.0	0.0000	0.0	0.0000	1.0000
3	2025-01-01 03:00:00	1.2092	0.0000	0.0000	1.2092	0.0	0.0000	0.0	0.0000	1.0000
4	2025-01-01 04:00:00	1.0283	0.0000	0.0000	1.0283	0.0	0.0000	0.0	0.0000	1.0000
5	2025-01-01 05:00:00	1.2927	0.0000	0.0000	1.2927	0.0	0.0000	0.0	0.0000	1.0000
6	2025-01-01 06:00:00	1.8806	0.0000	0.0000	1.8806	0.0	0.0000	0.0	0.0000	1.0000
7	2025-01-01 07:00:00	1.7252	0.9032	0.9032	0.8220	0.0	0.0000	0.0	0.0000	1.0000
8	2025-01-01 08:00:00	1.8219	1.4811	1.4811	0.3408	0.0	0.0000	0.0	0.0000	1.0000
9	2025-01-01 09:00:00	1.3909	1.8287	1.8287	0.0000	0.0	0.4378	0.0	0.4378	1.4159
10	2025-01-01 10:00:00	1.6311	2.5377	2.5377	0.0000	0.0	0.9066	0.0	0.9066	2.2772
11	2025-01-01 11:00:00	1.0709	2.1866	2.1866	0.0000	0.0	1.1156	0.0	1.1156	3.3370

Figure 1: Optimal hourly power flow (first 12 rows).

```
=== Battery charge/discharge events ===
```

	Timestamp	Battery_Charge_kw	Battery_Discharge_kw	SOC_kWh	Load_kw	Solar_kw	Grid_Import_kw	Grid_Export_kw
0	2025-01-01 09:00:00	0.4378	0.0000	1.4159	1.3909	1.8287	0.0000	0.0
1	2025-01-01 10:00:00	0.9066	0.0000	2.2772	1.6311	2.5377	0.0000	0.0
2	2025-01-01 11:00:00	1.1156	0.0000	3.3370	1.0709	2.1866	0.0000	0.0
3	2025-01-01 12:00:00	0.9329	0.0000	4.2233	1.1557	2.0886	0.0000	0.0
4	2025-01-01 13:00:00	1.1107	0.0000	5.2784	1.0088	2.1194	0.0000	0.0
5	2025-01-01 14:00:00	1.1353	0.0000	6.3570	1.1490	2.2843	0.0000	0.0
6	2025-01-01 15:00:00	0.6078	0.0000	6.9344	1.0741	1.6819	0.0000	0.0
7	2025-01-01 16:00:00	2.5211	0.0000	9.3294	1.3520	1.4090	2.4640	0.0
8	2025-01-01 17:00:00	0.0000	1.0480	8.2263	1.9340	0.8860	0.0000	0.0
9	2025-01-01 18:00:00	0.0000	1.9400	6.1842	1.9400	0.0000	0.0000	0.0
10	2025-01-01 19:00:00	0.0000	1.9610	4.1200	1.9610	0.0000	0.0000	0.0
11	2025-01-01 20:00:00	0.0000	1.8775	2.1437	1.8775	0.0000	0.0000	0.0
12	2025-01-01 21:00:00	0.0000	1.0865	1.0000	1.0865	0.0000	0.0000	0.0
13	2025-01-02 09:00:00	0.4154	0.0000	1.3946	1.5214	1.9368	0.0000	0.0

Figure 2: Battery charge and discharge events identified by the LP optimizer.

Result Analysis

Battery Charging/Discharging – LP Behaviour

The plot below shows how the LP optimizer decides when to charge and discharge the battery across the day. The model charges during high-solar, low-price hours and discharges during evening peak-price periods. This demonstrates intelligent, price-aware behaviour that the heuristic model cannot achieve.

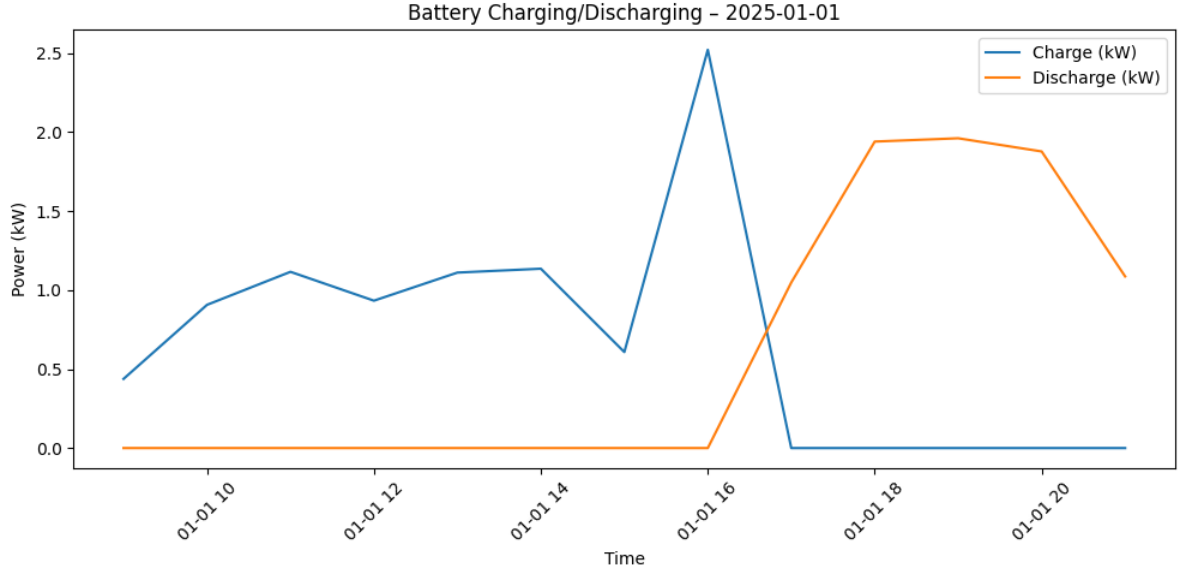


Figure 3: Battery charging and discharging profile – 2025-01-01.

State of Charge (SOC) Trend

The following plot illustrates how the battery's State of Charge evolves throughout the day. SOC increases during solar hours and decreases during evening demand hours, showing how the LP model manages energy storage in a smooth and optimal manner without violating SOC limits.

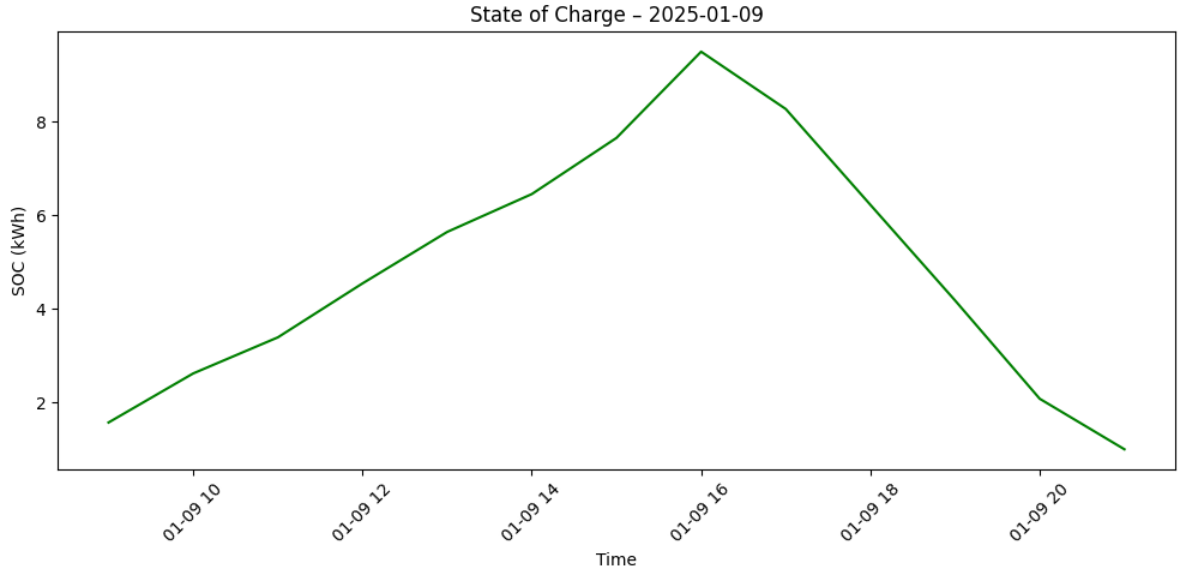


Figure 4: SOC variation across the day – 2025-01-09.

Power Flow Components Overview

The next plot visualizes the full energy flow for a representative day. It shows how solar generation is used to meet the load first, then to charge the battery, and finally exported if surplus exists. Grid power is imported only when essential. This confirms that the model coordinates solar, battery, and grid flows in an optimal manner.

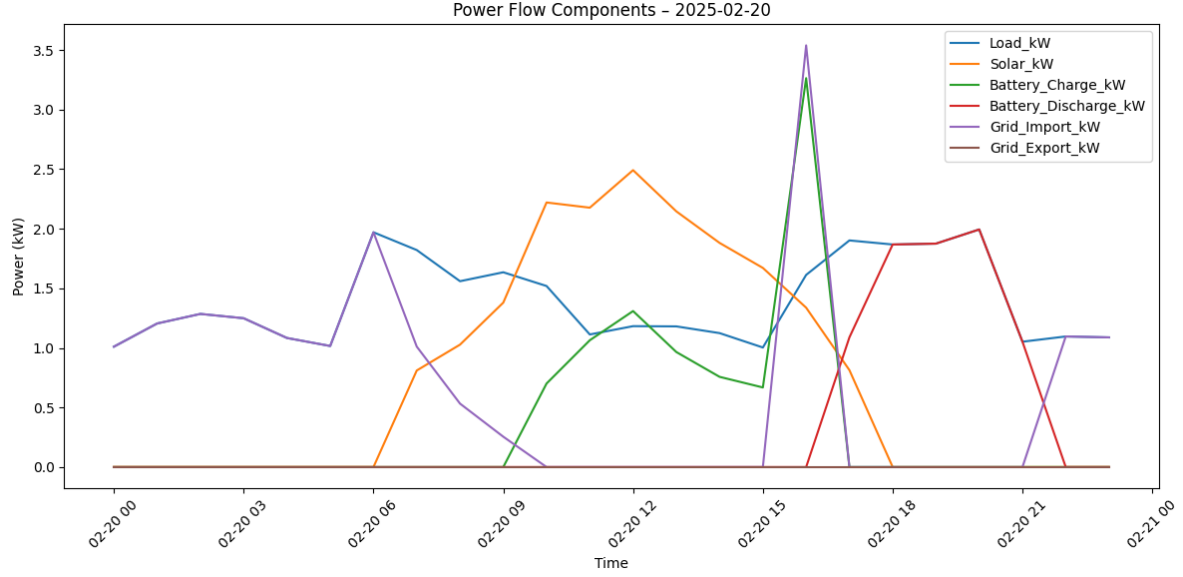


Figure 5: Power flow components – 2025-02-20.

Grid Import: LP vs Heuristic

This graph compares grid imports between the heuristic strategy and the LP model. The heuristic continues to import during high-price hours, whereas the LP model smartly avoids expensive grid consumption by discharging the battery. This demonstrates the clear cost advantage of optimization.

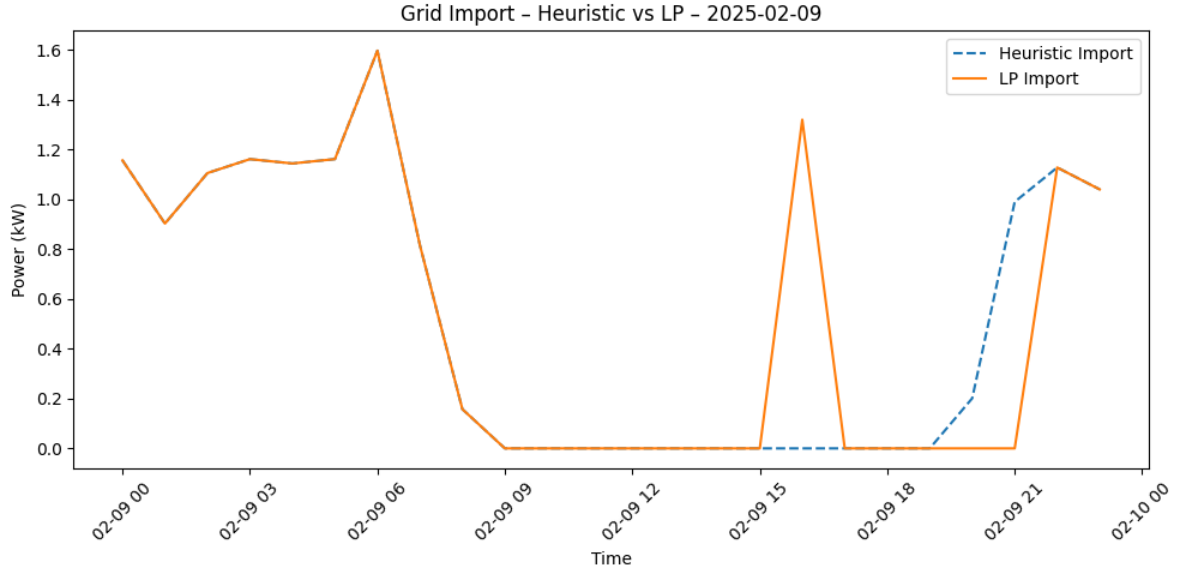


Figure 6: Grid import comparison – Heuristic vs LP – 2025-02-09.

Challenges and Fixes

During the development of the optimization model, several challenges were encountered. The key issues and the corresponding fixes are summarized below:

1. Battery Overflow During High Solar Generation

Challenge: Without proper bounds, the optimizer tended to overcharge the battery during hours of high solar output, leading to unrealistic or physically impossible SOC values.

Fix Implemented: Explicit SOC limits were enforced:

$$SOC_{\min} \leq E_t \leq SOC_{\max}$$

along with charge/discharge power bounds:

$$0 \leq P_t^{ch} \leq P_{\max}^{ch}, \quad 0 \leq P_t^{dis} \leq P_{\max}^{dis}.$$

These constraints ensured that the battery never exceeded its physical capacity even during high solar production periods.

2. CVXPY Compatibility Issues

Challenge: The initial implementation attempted to use CVXPY, but solver availability and installation constraints (especially on student systems and during deployment) caused compatibility issues.

Fix Implemented: The model was rewritten using **PuLP**, a lightweight and solver-agnostic Python LP library. PuLP integrates smoothly with CBC (default solver), requires no complex setup, and is ideal for linear programs like this one.

3. Penalizing Unnecessary Battery Cycling

Challenge: Early versions of the model caused excessive charge–discharge cycling, which is unrealistic and leads to higher long-term battery degradation.

Fix Implemented: A penalty term was added to the objective:

$$k (P_t^{ch} + P_t^{dis})$$

where

$$k = \text{DEGRAD_COST} + \text{CH_DIS_PENALTY}.$$

This discouraged unnecessary cycling, reduced battery stress, and resulted in smoother, more realistic battery usage patterns.